

Two-layer perceptron from scratch

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1 Inputs

$$\mathbf{X} \in \mathbb{R}^{N \times D}, \mathbf{y} \in \mathbb{R}^{N \times C}$$

2 The Network

$$\mathbf{W}_1 \in \mathbb{R}^{D \times H}, \mathbf{b}_1 \in \mathbb{R}^{1 \times H}$$

$$\mathbf{W}_2 \in \mathbb{R}^{H \times C}, \mathbf{b}_2 \in \mathbb{R}^{1 \times C}$$

$$\mathbf{h}_1 = \mathbf{X}\mathbf{W}_1 + \mathbf{b}_1 \in \mathbb{R}^{N \times H}$$

$$\mathbf{h}_2 = f(\mathbf{h}_1) \in \mathbb{R}^{N \times H}$$

$$\mathbf{h}_3 = \mathbf{h}_2\mathbf{W}_2 + \mathbf{b}_2 \in \mathbb{R}^{N \times C}$$

Note: Below, we will use the notations $\mathbf{h}_i$ and $\mathbf{h}^{(i)}$ interchangeably so as to make the text more legible. In particular, when it's necessary to index into $\mathbf{h}_i$, we will use the $\mathbf{h}_j^{(i)}$ notation to refer to the j th element in $\mathbf{h}_i$.

3 Nonlinearity

$$\begin{aligned} f(\mathbf{X}) = \text{ReLU}(\mathbf{X}) &= \begin{bmatrix} \text{ReLU}(x_{11}) & \cdots & \text{ReLU}(x_{1j}) & \cdots & \text{ReLU}(x_{1n}) \\ \vdots & & \ddots & & \vdots \\ & & \text{ReLU}(x_{ij}) & & \\ \vdots & & & \ddots & \vdots \\ \text{ReLU}(x_{m1}) & \cdots & \text{ReLU}(x_{mj}) & \cdots & \text{ReLU}(x_{mn}) \end{bmatrix} \\ &= \begin{cases} x_{ij} & , \text{ if } x_{ij} \geq 0 \\ 0 & , \text{ if } x_{ij} < 0 \end{cases} \end{aligned}$$

4 Predictions

$$\mathbf{q}_i = \text{softmax}(\mathbf{h}_3)_i$$

$$\text{softmax}(\mathbf{z})_i = \text{softmax}(\mathbf{z}_i) = \frac{e^{\mathbf{z}_i}}{\sum_{k=1}^C e^{\mathbf{z}_k}}$$

5 Loss

Below, i is the index of the correct class.

$$\begin{aligned} \mathcal{L} &= \text{CE}(\underbrace{p}_{\text{true}}, \underbrace{q}_{\text{estimate}}) = -\mathbb{E}_p[\log q] \\ &= -\sum_{k=1}^C \mathbf{p}_k \log \mathbf{q}_k \\ &= -\cancel{\mathbf{p}_i}^1 \cdot \log\left(\frac{e^{\mathbf{h}_i^{(3)}}}{\sum_{k=1}^C e^{\mathbf{h}_k^{(3)}}}\right) \\ &= -\log e^{\mathbf{h}_i^{(3)}} + \log\left(\sum_{k=1}^C e^{\mathbf{h}_k^{(3)}}\right) \\ &= \boxed{\log\left(\sum_{k=1}^C e^{\mathbf{h}_k^{(3)}}\right) - \mathbf{h}_i^{(3)}} \end{aligned}$$

6 Gradients (backprop)

Outputs:

$$\nabla_{\mathbf{h}_i^{(3)}} \mathcal{L}_i = \begin{cases} \frac{1}{\sum_k e^{\mathbf{h}_k^{(3)}}} e^{\mathbf{h}_i^{(3)}} - 1 & , i \text{ is correct class} \\ \mathbf{q}_i + 0 & , i \text{ is incorrect class} \end{cases}$$

$$\boxed{\nabla_{\mathbf{h}_i^{(3)}} \mathcal{L} = \mathbf{q} - \mathbf{p}} \in \mathbb{R}^{N \times C}$$

Layer 2:

$$\begin{aligned}\frac{\partial \mathbf{h}_3}{\partial \mathbf{W}_2} &= \mathbf{h}_2^\top \Rightarrow \frac{\partial \mathcal{L}}{\partial \mathbf{W}_2} = \frac{\partial \mathbf{h}_3}{\partial \mathbf{W}_2} \cdot \frac{\partial \mathcal{L}}{\partial \mathbf{h}_3} \\ &\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{W}_2} = \mathbf{h}_2^\top (\mathbf{q} - \mathbf{p})} \in \mathbb{R}^{H \times C}\end{aligned}$$

$$\frac{\partial \mathbf{h}_3}{\partial \mathbf{b}_2} = 1 \Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{b}_2} = \mathbf{q} - \mathbf{p}} \in \mathbb{R}^{N \times C}$$

$$\frac{\partial \mathbf{h}_3}{\partial \mathbf{h}_2} = \mathbf{W}_2^\top \Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_2} = (\mathbf{q} - \mathbf{p}) \mathbf{W}_2^\top} \in \mathbb{R}^{N \times H}$$

Layer 1 nonlinearity:

$$\begin{aligned}\frac{\partial \mathbf{h}_2}{\partial \mathbf{h}_1} &= \nabla \text{ReLU} = \begin{cases} +1 & \mathbf{h}_{ij}^{(1)} \geq 0 \\ 0 & \mathbf{h}_{ij}^{(1)} < 0 \end{cases} = \mathbb{1}(\mathbf{h}_{ij}^{(1)}) \\ &\Rightarrow \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_2} \frac{\partial \mathbf{h}_2}{\partial \mathbf{h}_1} \\ &\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{h}_1} = \mathbb{1}\left(\frac{\partial \mathcal{L}}{\partial \mathbf{h}_2}\right)} \in \mathbb{R}^{N \times H}\end{aligned}$$

Layer 1:

$$\begin{aligned}\frac{\partial \mathbf{h}_1}{\partial \mathbf{W}_1} &= \mathbf{X}^\top \\ &\Rightarrow \frac{\partial \mathcal{L}}{\partial \mathbf{W}_1} = \frac{\partial \mathbf{h}_1}{\partial \mathbf{W}_1} \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1} \\ &\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{W}_1} = \mathbf{X}^\top \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1}} \in \mathbb{R}^{D \times H}\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathbf{h}_1}{\partial \mathbf{b}_1} &= 1 \\ &\Rightarrow \boxed{\frac{\partial \mathcal{L}}{\partial \mathbf{b}_1} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}_1}} \in \mathbb{R}^{N \times H}\end{aligned}$$